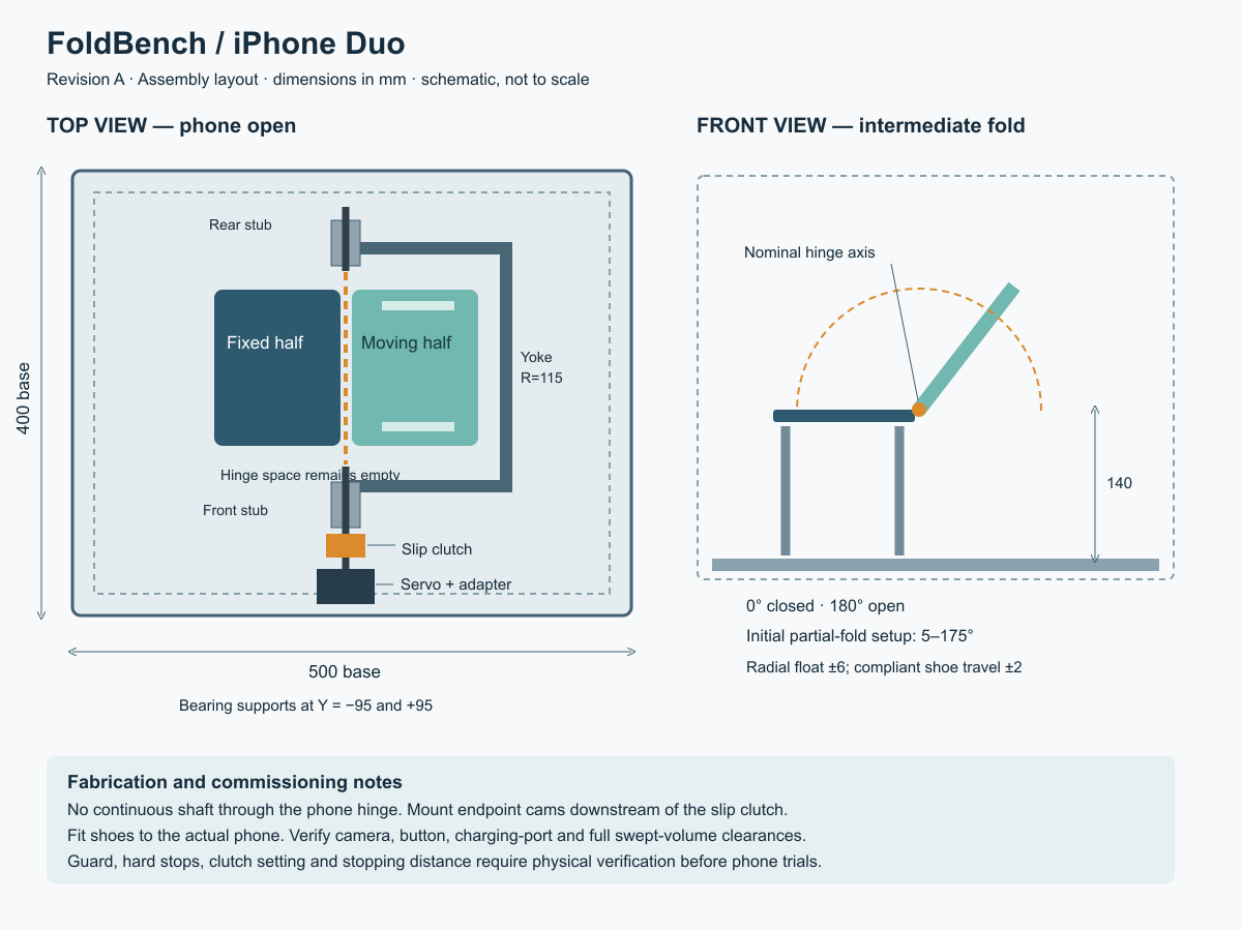
FOLDBENCH

iPhone Duo endurance tester

Design, procurement, assembly and Raspberry Pi control

Revision A / 4 November 2026

A guarded single-phone folding fixture with current-limited actuation, independent endpoint verification, durable cycle records and an authenticated FastAPI interface. Includes a simulator and real hardware driver.



Station parts allowance: $2,510; phone and fabrication labor excluded. Software checked locally; hardware unbuilt and unvalidated. Commission before any phone trial.

FoldBench / iPhone Duo

Revision A - prototype engineering package - 4 November 2026

1. What is being built

A single-station, guarded benchtop folding-endurance tester. A Raspberry Pi runs Python/FastAPI and commands a smart servo through USB. One phone half is supported in a stationary cradle; the other is carried by a rotating, floating cradle. The machine records verified open-close-open cycles, motor current, temperatures, faults and operator inspections.

This is an original design inspired by CNET's machine, not a reverse-engineered copy or a claim to reproduce its test method. The supplied Python software is executable and simulator-tested. The mechanics, wiring, Pi GPIO integration, stopping time, continuous-duty capability and phone fit have not been physically validated. Treat this package as a prototype for fabrication and commissioning, not a certified unattended test machine.

Video observations versus design decisions

The linked 60-second CNET clip shows an aluminum-framed mechanism repeatedly opening and closing a phone, a moving cradle/linkage, cables, a red stop button and a displayed cycle counter. The caption identifies the phone as iPhone Duo and describes continuous operation. The clip does not establish a motor model, torque, speed profile, hinge trajectory, sensor architecture or dimensioned drawings. Those details below are our design choices. Playback and several mechanism views were inspected; no audio transcript was available.

Apple lists the Duo at 164.6 x 117.8 x 5.2 mm open, 84.1 x 117.8 x 11.3 mm closed, and 254 g. These are exterior envelopes, not a fixture drawing. Measure the actual phone's camera protrusion, half-body widths, buttons, port location, edge radii, hinge migration and usable contact patches before machining the phone-contact shoes. [S1, S2]

Prototype operating contract

Parameter	Revision A choice
Specimen	One iPhone Duo, case removed; USB-port half stationary
Base / outer guard envelope	500 x 400 mm / approximately 600 x 500 x 400 mm
Motion	One motor, 1:1 shaft drive, horizontal hinge axis, floating moving cradle
Angle convention	0 degrees closed, 180 degrees open
Initial test span	5-175 degrees; partial-fold screening, explicitly recorded
Full closure	Only after phone-specific fit and force qualification; use measured angles/ticks
Speed	1-30 degrees/s request range; initial installation ceiling 15 degrees/s
Endpoint dwell	0.25-60 seconds; default 1 second
Current	Default ceiling 150 mA; commissioning range 20-350 mA
Mechanical torque limiter	Initially 0.20 N m; adjust only from measured baseline, maximum design target 0.50 N m
Sampling	Nominal 20 Hz control loop; stored 1 Hz plus confirmed endpoint samples
Cycle definition	Confirmed open, then closed, then open; stable endpoint evidence for 150 ms
Persistence	SQLite FULL synchronous commits; one commit per counted cycle
Restart	No automatic motion or run resumption
Inspection pause	Every 1,000 cycles by default; operator must record pass to resume
Phone / ambient / motor limits	Initial case 40 C, ambient 10-30 C, motor 55 C

These limits are conservative starting choices, not measured safe limits for the Duo. Motor current is not calibrated hinge torque. A case sensor does not measure internal battery temperature. A completed run means the requested mechanical cycling finished; it does not mean the phone passed functional testing.

Ignoring acceleration, the default 170-degree span gives a cycle time of 2 x 170 / 15 + 2 x 1 = 24.7 seconds, or at most about 3,500 cycles/day. Acceleration, verification and inspections reduce throughput. Do not compare this directly with CNET's cycle totals or convert cycles into years of normal use.

2. Architecture and interlocks

The Pi has its own 27 W supply and remains powered after an emergency stop. An enclosed 12 V, 5 A supply feeds the motor through a fused safety-relay output path. A separate enclosed 24 V supply feeds the safety relay. No exposed mains wiring is required inside the machine.

Use a Pilz PNOZ X3 774310 with a two-contact emergency stop and a two-contact tongue-operated guard switch. Both safety channels must open when either device is operated. Use monitored physical reset, outside the guard. Route the motor's positive supply through two safety output contacts in series. This removes motor power independently of Python, Linux and network availability. The third safety output drives an isolated Pi status input. The exact relay input wiring must follow the manufacturer's dual-channel, cross-short-detection diagram; it must be checked against the supplied revision before energizing. No safety performance level is claimed for the assembled prototype. [S5]

Normal API stop disables torque. API protective stop latches a fault. Neither guarantees an immediate physical stop. Power removal can allow the cradle to settle under gravity: retain the phone mechanically, damp the moving cradle, and measure the worst-case coast/fall motion during commissioning. Place the guard switch so motion stops before a hand can reach the swept volume; use a locking guard if the measured time does not permit this.

The actuator bus watchdog stops motion after loss of command traffic; it can retain holding torque and is not the emergency-stop circuit. Mode 5 provides current-based position control but does not enforce the servo's min/max position registers. Independent adjustable fixture stops and the slip clutch are therefore essential. [S3]

3. Mechanical layout

Use the supplied layout SVG and parametric OpenSCAD source as design references. The source models the assembly envelope and provides flat stock/slot templates. Vendor bearing, horn, slide and motor mounting patterns must be taken from the parts actually delivered and transfer-drilled into the slotted adapter plates. The model is not a collision-certified or tolerance-validated production assembly.

Coordinate system: origin at the center of the base's top surface. X is left/right across the phone, Y follows the phone hinge, Z is upward. The nominal phone hinge axis is X=0, Z=140 mm. The phone height runs from Y=-58.9 to +58.9 mm. The stationary half is at negative X; the moving half is at positive X when open. The front of the machine is negative Y.

The hinge must remain unobstructed

Use two separate 6 mm stub shafts, one in front of and one behind the phone. Do not install a continuous shaft through the phone hinge. Front/rear support centers are Y=-95 and +95 mm. The front stub projects toward the motor; the rear stub is an idler. Bearings and drive components stay outside Y=+/-70 mm. A U-shaped rotating yoke joins the two stubs through two radial arms and an outer crossbar at X=115 mm. This leaves the phone hinge space empty.

Each half is supported from its exterior back, with soft shoes on suitable metal-frame areas. Nothing presses on the flexible display, hinge gap, buttons, camera glass or USB connector. The moving plate rides on two radial linear guides, allowing +/-6 mm translation; compliant shoe mounts allow approximately +/-2 mm tangential movement. The phone's real hinge determines the path. It must not be forced to behave like a perfect pin hinge.

Fabricated parts and dimensions

All dimensions in mm. General noncritical dimensions +/-0.5; hole locations +/-0.2; bearing fits and shaft sizes to the purchased bearing specification. Deburr all edges, minimum 0.5 edge break. Phone-adjacent corners radius at least 2. Use 6061-T6 aluminum for structural plates; acetal for shoe bodies; 1 mm silicone pads, approximately Shore A 30-40, for replaceable contact surfaces.

Drawing ID	Qty	Stock and operations
M01 base	1	500 x 400 x 8 aluminum; mounting grid holes 6.6 diameter on 25 mm grid in center 350 x 250 area; four foot holes at 20 mm from corners
M02 bearing towers	2	100 x 180 x 8 aluminum, 100 x 50 x 6 angle feet; slotted bearing adapter permits Z=130-150; tower feet permit X +/-10 and Y +/-15
M03 yoke radial arms	2	125 x 25 x 6 aluminum; 6 mm split-clamp hub at inner end; crossbar holes at X=115, two M4 fixings per joint
M04 yoke outer crossbar	1	190 x 25 x 6 aluminum; bridges arms at Y +/-95; keep outside phone edge

Drawing ID	Qty	Stock and operations
M05 moving backplate	1	100 x 135 x 4 aluminum; four 4.5 x 20 adjustment slots; attach to slide carriages, not rigidly to yoke
M06 fixed backplate	1	100 x 135 x 4 aluminum; camera clearance pocket located from actual phone; four 4.5 x 20 adjustment slots
M07 fixed cradle risers	2	100 x 125 x 6 aluminum vertical plates plus 100 x 40 x 6 feet; final height shimmed to phone exterior back
M08 shoe bodies	8	Acetal blanks 20 x 15 x 8; slot 4.5 x 10; machine contact contour after specimen measurement; four shoes per half
M09 motor shelf	1	100 x 100 x 6 aluminum on two 100 mm slotted angles; position motor axis coaxial with front shaft at Z=140
M10 stop/sensor mounts	4	40 x 25 x 3 aluminum angles; slotted base holes 4.5 x 15; two for stops and two for switches
M11 horn-to-shaft adapter	1	30 diameter x 10 aluminum hub; 6 H7 bore with split clamp; horn fixing pattern matched to supplied ROBOTIS horn; no invented spline dimensions
M12 guide carrier brackets	2	100 x 25 x 4 aluminum; slotted holes for the two MGN7 guide rails; attach across yoke arms behind moving plate
M13 switch cams	2	20 x 15 x 3 acetal; bolt to moving cradle/yoke output side of clutch; adjustable trigger angle
M14 guard panels	5	4 mm polycarbonate: front/rear 560 x 360 (2), sides 460 x 360 (2), top 560 x 460 (1); mounting overlaps adjusted to frame

The fixed plate's back surface orientation and camera cutout differ between phone halves; do not make mirror-image contact shoes without checking the specimen. Use pads on the rigid perimeter only. The clamp restraint must resist a 10 N tangential pull without crushing or changing the manual hinge feel. If that cannot be achieved on a non-screen contact patch, redesign the shoes rather than tightening them further.

The free carriage travel must remain at least 2 mm from either mechanical travel limit at every angle. End stops act on the yoke, not the phone. For the initial 5-175-degree trial, set stops approximately 1 degree outside the software endpoints, still within the specimen's freely moving range. For a full-range test, the stops must be established from actual safe closure/opening and must never force negative opening or overextension.

Drive sizing and limitations

At an assumed moving phone-half mass of 0.127 kg and center radius 0.042 m, gravity contributes about 0.052 N m. A 0.15 kg lightweight moving fixture at radius 0.05 m adds about 0.074 N m. Together these are roughly 0.126 N m before hinge friction, acceleration and imbalance. Weigh the finished fixture and measure the real force. This estimate is why an unqualified low-current setting may stall.

The XM430 has ample peak torque, but peak/stall torque is not a continuous-duty rating. Do not use its full rating on a phone. Set the friction clutch above the measured normal folding torque with a small documented margin and below the established specimen/fixture limit, never above this prototype's 0.50 N m design ceiling. At a 60 mm lever, 0.20 N m corresponds to 3.33 N tangential force. If baseline folding requires more than the available protective range, stop and revise the fixture or counterbalance; do not override limits.

For the clutch, specify Dynatect Polyclutch EAS12, shaft-to-shaft, 6 mm input/output bores (configured request EAS12-6MM-6MM). Obtain a vendor quote confirming bidirectional low-speed oscillation and adjustment around 0.20-0.50 N m. This is a configured item, not a confirmed in-stock SKU. Its published nominal envelope is 19.05 diameter x 31.75 long; our design uses a horizontal axis as required by the installation instructions. Require supplier confirmation of low-end adjustment and breakaway torque before ordering. [S6]

4. Bill of materials

The complete line-item bill is supplied as BOM.json and BOM.md, including stock, fasteners, wiring, software, calibration tools and optional observation equipment. Quantities are for one station. Prices are planning allowances in USD, not supplier quotations. The phone, taxes, shipping, professional fabrication labor and a suitable test-room fire response arrangement are excluded. The clutch configuration and phone shoes require final fit/quote confirmation.

Use equivalent parts only if the interfaces, electrical ratings, failure behavior and mounting requirements are rechecked. Do not substitute an unisolated hobby relay board for the safety relay, or a 5 V logic connection for a 3.3 V Pi input.

5. Electrical connections

Power and relay output netlist

Net	Connection
P1	Pi official 27 W USB-C supply directly to Pi; independent of motor stop circuit
P2	Enclosed 12 V 5 A supply + -> DC disconnect -> 3 A fast fuse F1 -> safety relay 13
P3	Relay 14 -> relay 23; relay 24 -> U2D2 power-hub +12 V input
P4	12 V supply negative -> hub motor ground; use ROBOTIS keyed cables to servo
P5	Pi USB -> U2D2; U2D2 TTL 3-pin port -> hub -> XM430-W350-T; do not use the RS-485 port
P6	Enclosed 24 V 1 A supply + -> 0.5 A fuse F2 -> PNOZ B1; 24 V negative -> B2
P7	24 V fused + -> relay 33; relay 34 -> 2.0 kohm 0.5 W resistor -> optocoupler LED anode; cathode -> 24 V negative
P8	Optocoupler collector -> Pi 3.3 V; emitter -> GPIO17 node; 10 kohm from node to Pi ground; 1 kohm series from node to GPIO17 header pin
P9	Pi GPIO inputs use Pi ground only; no 24 V conductor connects directly to any Pi pin

Use an LTV-817C or equivalent optocoupler with documented CTR at the chosen current; its LED receives approximately 11 mA from the 24 V status circuit. Confirm GPIO high is above 2.0 V and below 3.3 V. Check the selected part's pinout. Add a reverse-protection diode across the optocoupler LED. The 33-34 circuit is status only. Power-cut testing at the actual motor terminals is still required.

Use the manufacturer's two-channel E-stop/safety-gate circuit with cross-short detection and monitored reset. Put the two E-stop NC contacts and the corresponding guard NC contacts in their separate channels. For PNOZ X3 DC supply use B1/B2, not an assumed A1/A2 connection. Monitored reset uses S33/S34; leave automatic-start S13/S14 unlinked. Preserve or remove the delivered S11/S12 link only as the selected vendor diagram specifies. The manual's schematic, rather than a guessed text pin mapping, is the authority for the input circuit. Check fault masking in the

series-connected devices and validate each device independently. [S5]

Pi header table (BCM numbering)

Signal	BCM line	Physical header pin	Connection
Safety permit	17	11	Isolated active-high status from P8
Open endpoint	27	13	Gold-contact switch NO to 3.3 V; 10 kohm pulldown; 1 kohm series to Pi
Closed endpoint	22	15	Same circuit as open switch
1-Wire data	4	7	Both DS18B20 data wires; one 4.7 kohm pullup to 3.3 V
3.3 V	-	1	Probe supply and input circuit supply only
Ground	-	6	Probe/input ground

Use three-wire powered temperature probes, not parasite power. Identify their actual 28-* serial numbers. Attach the case probe using thin removable thermal tape to the stationary half's exterior frame/back, away from the camera, not to the folding display. Place the ambient probe in shaded, freely circulating air near the phone. The case probe must remain attached through all folds; its loss of thermal contact is not automatically detectable from a valid digital reading.

Use 18 AWG for 12 V distribution, 22 AWG for safety wiring, 24-26 AWG twisted pairs for short signal runs. Use ferrules, numbered terminals, strain relief and covered connections. Keep signal wiring away from the clutch/shaft and motor cabling. The USB charging cable, if used, goes to the stationary phone half with a loose strain-relieved loop. Continuous charging is a separate test condition: initially run uncharged and periodically inspect the battery/phone. A case-temperature stop does not certify battery safety.

6. Assembly sequence

1. Inventory all BOM lines. Print the drawings at labeled scale only when explicitly shown; the overview is not a drilling template. Verify vendor dimensions and clutch configuration before cutting adapter parts.

2. Machine and deburr the base, towers, yoke, plate blanks and slotted brackets. Bolt on the four feet. Support the wider guard frame on fixed outriggers from the base plate. Assemble the base frame square; measure diagonals within 1 mm. Do not install the phone.

3. Mount the bearing towers at Y=+/-95. Use a temporary straight alignment rod through the two bearing bores to align them at X=0, Z=140 within 0.2 mm. Remove this rod: it must never be present with the phone installed.

4. Install the two short stub shafts, radial yoke arms and outer crossbar. Use two shaft collars per support to restrain axial motion. Verify the empty yoke rotates without binding. A shaft must not project into the phone envelope.

5. Mount the motor on its slotted shelf outside the front support. Connect the verified horn adapter, short input shaft and slip clutch. Align the clutch axes within 0.1 mm as an assembly target. No phone loads should be carried by the servo output bearing alone. Tighten hub screws to the hub supplier's value; apply witness marks.

6. Mount the two radial rails and moving plate. Confirm +/-6 mm available travel, soft retention at travel extremes, and no loose ball carriage. Fit the four compliant shoe mounts. Adjust float without appreciable preload on a dummy phone.

7. Build the fixed cradle and align the two phone backing planes using a two-leaf dummy of the measured phone envelope. Include camera/connector relief before any real-phone fit. Ensure the yoke and brackets clear both halves over the entire arc.

8. Fit adjustable stops and the two endpoint switches. Switch cams must be on the output side of the clutch. Initial switching should occur close to the calibrated endpoint, never throughout a large part of the stroke. Fit locknuts and witness marks.

9. Assemble the polycarbonate guard and front door. Fit the captive tongue switch and E-stop/reset panel. Leave at least 25 mm clearance between guard and every moving part. Avoid openings through which fingers can reach the mechanism; assess ventilation slots and cable entries for reach-through.

10. Install the low-voltage electrical enclosure, DIN rail, relay, fuses, DC disconnect, terminal blocks and isolated input board. Connect exactly as the netlist and vendor relay diagram require. Label the 12 V, 24 V and 3.3 V domains distinctly.

11. Fit the Pi, cooler and storage outside the motion envelope. Connect USB and probes; route all wires clear of stops, rails and pinch points. Check continuity and polarity before connecting power supplies.

12. With the servo disconnected, test guard and E-stop operation using a meter at the hub output. Confirm physical reset is required after each interruption, that holding reset cannot defeat the intended monitored behavior, and that reset never commands motor motion.

7. Commissioning and calibration

Record results in the supplied commissioning checklist. Never set `commissioned=true` based solely on the example file. The API cannot remotely widen installation limits.

A. Verify electronics with motor uncoupled

Set servo ID 1, Protocol 2, 57600 baud. These are the expected XM430 settings. Stop the service and use `pipenv run python -m foldbench.probe --port /dev/serial/by-id/...` to read identity and position with torque disabled. The probe utility does not command movement. Verify model 1020 and firmware at least 38. If ID/baud differs, restore the documented configuration using the vendor's setup procedure before connecting the application. The application does not scan or broadcast to unknown actuators.

First set up an empty mechanism. With the motor mechanically disconnected, set the yoke at the chosen closed angle against an angle gauge. Select an encoder region at least 256 counts from rollover, with room for about 2,000 counts of positive movement. The supplied Python setup command can index the mechanically uncoupled motor: append `--index-tick 1024 --uncoupled --gpio-chip /dev/gpiochip0` to the probe command, using the verified GPIO chip. This enables a roughly 48 mA, 4-degree/s setup move only while the guard/safety permit is healthy, then disables torque. The flag is an operator declaration: the software cannot verify that the coupling and phone are absent. Never use this command on a coupled mechanism. Record the observed tick, reconnect the clutch without moving either side, and use angle-span x 4096/360 for the preliminary open-tick calculation. Check positive motor direction opens the phone; reverse mechanical orientation if necessary, since this implementation expects open_tick > closed_tick.

The Python operational controller and uncoupled indexing utility are supplied. A servo with a nondefault ID/baud may require vendor recovery before use. No handset-control software is required. No claim is made that iOS exposes display health, touch health or internal battery temperature through this REST API.

B. Empty mechanism and dummy qualification

1. Verify GPIO chip identity from `/sys/bus/gpio/devices/gpiochip*/label`. Select the chip that owns the header BCM lines. Chip numbers differ between Pi/kernel versions. The driver uses the Linux GPIO v1 character ABI; if the kernel disables this ABI, stop and adapt/test the driver before use.

2. Enable the `w1-gpio` overlay on BCM4 in the Pi boot configuration. Verify two distinct probes produce CRC-valid readings. Identify case versus ambient by gently warming one and record the serial mapping.

3. Verify inputs at the API or with a meter: healthy permit high, each endpoint high only at its own position, open wires low. Both endpoints must never be high together.

4. Inspect all hard stops, guard wiring, clutch alignment and dummy clearance. Create an EMPTY-FIXTURE commissioning record and a temporary calibration file for dummy-only testing, with 5 degrees/s and the lowest current capable of moving the empty fixture. Do not reuse this record for a phone test.

5. Run single endpoint jogs with the guard closed, no phone, and a hand at the E-stop. Check measured direction and travel. Operate E-stop and open the guard independently during motion. Measure stopping time and distance, including gravity settling; record the worst value and verify access is prevented until motion stops.

6. Disconnect USB during movement. The bus watchdog must trip, and reconnect must not resume. The resulting watchdog fault requires inspection and a motor power cycle before reset. Stop the Pi process and cut Pi power separately; verify no automatic restart.

7. Jam the dummy gently with a calibrated restraint, not fingers. Verify clutch slips below its qualified torque, endpoint timeout faults, the count does not increase, and power interruption releases drive. Test a disconnected endpoint wire and each disconnected temperature probe.

8. Run 100 dummy cycles, recheck alignment and clamp marks, then run 1,000 dummy cycles while checking motor temperature, rails, cable wear and counts. Reconcile the machine count against an independent tally/video sample.

C. Phone-specific qualification

Measure the actual Duo and complete its shoes. Support the phone with torque off and clutch disengaged. Move it by hand over the entire intended range; there must be no forced translation, clamp movement, screen contact or cable tug. Record natural open/closed angles and the radial slide excursion at 0, 45, 90, 135 and 180 degrees where physically reachable.

Measure the peak folding force gently at a known lever radius, in both directions and at several angles. Include empty-fixture gravity/friction separately. Set the clutch using a force gauge on a 100 mm arm: torque in N m = force in N x 0.100. Test clockwise and counterclockwise breakaway repeatedly; require repeatability within the documented acceptance band, initially +/-10%. Lock and mark the adjuster.

Set current only high enough to complete the stroke with margin below the measured mechanical protection threshold. The firmware current scale is not a torque calibration. Record the relationship from actual measurements. If the phone cannot move within the 350 mA / 0.50 N m ceiling, revise the mechanics; do not silently raise it.

Start with 5-175-degree partial folds at 5 degrees/s, 2-second dwells, one cycle. Inspect the phone, pads and cable. Repeat ten cycles, then 100. Only after these pass may the phone-specific `commissioned=true` record authorize routine runs. Full 0-180-degree operation needs a new measurement/stop/switch setup and recorded calibration; use whatever real angles the phone freely reaches, not forced mathematical endpoints.

D. Run acceptance and result interpretation

Record baseline photographs, serial identifier, iOS version, battery state, charge state, room temperature, fixture revision and pre-existing damage. At each scheduled pause inspect display uniformity, crease, touch response, frame, hinge feel, clamp witness marks and USB cable. Record pass/fail and notes through the API. The software cannot judge these automatically. A fail ends the run in a fault state; a pass permits, but does not initiate, resume.

Power loss can occur after a physical cycle but before its database commit. The persisted count is a confirmed lower bound and may omit that last cycle. Partial strokes are deliberately uncounted. Do not merge counts across interrupted/replaced specimens without an explicit analysis record.

8. Maintenance and unattended operation boundary

Before each run: operate E-stop and guard, inspect fasteners/pads/cable, check probe attachment, disk space and serial/time settings. After each 1,000 cycles initially: check clutch setting, witness marks and rails; set the later interval from measured wear. Lubricate only the rail/bearing locations specified by their manufacturers. Keep lubricant away from the clutch and phone. Recommission after changing shoes, clutch, firmware, servo, stops, sensors or safety wiring.

Revision A is intended for supervised endurance testing. A 24/7 unattended release needs physical validation under the intended room conditions, independent battery/thermal incident provisions, long-duration component qualification, a completed machine risk assessment and confirmed stopping performance. The video demonstrates an operating aim, not evidence that this newly designed machine is ready for that service.

9. Software and API behavior

See README.md and docs/API.md for installation and requests. FastAPI serves interactive documentation at `/docs`, the live schema at `/openapi.json`, and the same exported schema in openapi.json. All operational endpoints require `X-API-Key`. Start requests also require `Idempotency-Key`; retries return the original run and changing the body with the same key is rejected.

Only one run or jog may own the mechanism. A dedicated worker serializes hardware commands. Temperature conversion runs independently so it cannot block bus-watchdog servicing. The process locks its data directory and exclusively opens the serial port. Do not use Uvicorn reload, multiple workers or multiple services with different data directories on the same station.

Results include the exact recipe and commissioning snapshot, simulated/physical mode, phase, confirmed cycles, timestamps and fault. Samples include motor-derived angle, signed current, motor/case/ambient temperatures, supply voltage and endpoint/permit states. Samples and events use bounded pagination; CSV export returns an `X-Next-After` cursor. The software does not auto-delete test data. It stops a running test when free space falls below 100 MiB.

If torque-off cannot be confirmed, the controller reports that explicitly and stops background bus polling so it cannot keep feeding the watchdog. Use the physical E-stop. Any API stop acknowledgement means the request was accepted, not that a safety-rated stop was proven.

10. Sources and evidence

Sources checked 4 November 2026. Component pricing in the BOM is an estimate independent of source-listed retail prices.

- S1: CNET video/post: https://x.com/CNET/status/2114060888109928499
- S2: Apple Duo exterior dimensions: https://www.apple.com/iphone-duo/specs/

• S3: ROBOTIS XM430-W350 control table, operating modes and watchdog: https://emanual.robotis.com/docs/en/dxl/x/xm430-w350/

• S4: ROBOTIS U2D2 connections: https://emanual.robotis.com/docs/en/parts/interface/u2d2/

• S5: Pilz PNOZ X3 manual, especially wiring and preparation pages 11-14: https://www.pilz.com/download/open/PNOZ_X3_Operating_Manual_20547-EN-10.pdf

• S6: Dynatect Slip-Ease specifications and horizontal installation: https://www.dynatect.de/en/product/slip-ease-slip-clutches/

• S7: Raspberry Pi power guidance: https://www.raspberrypi.com/documentation/computers/getting-started.html

• S8: OMRON D4NS guard switch: https://industrial.omron.co.uk/en/products/D4NS-1BF

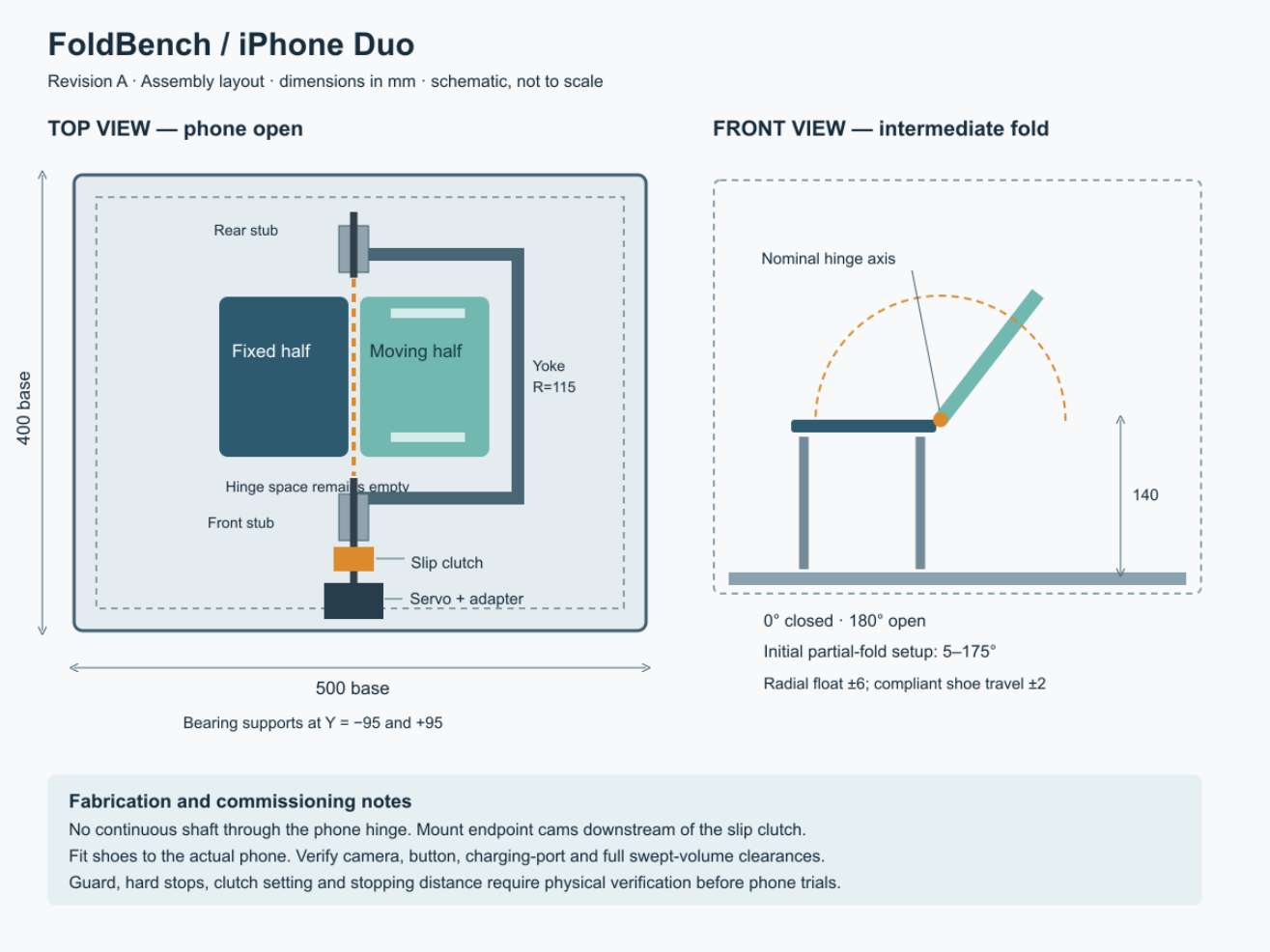
• S9: FastAPI lifespan: https://fastapi.tiangolo.com/advanced/events/

• S10: FastAPI API key security: https://fastapi.tiangolo.com/reference/security/

• S11: DYNAMIXEL Python SDK: https://github.com/ROBOTIS-GIT/DynamixelSDK

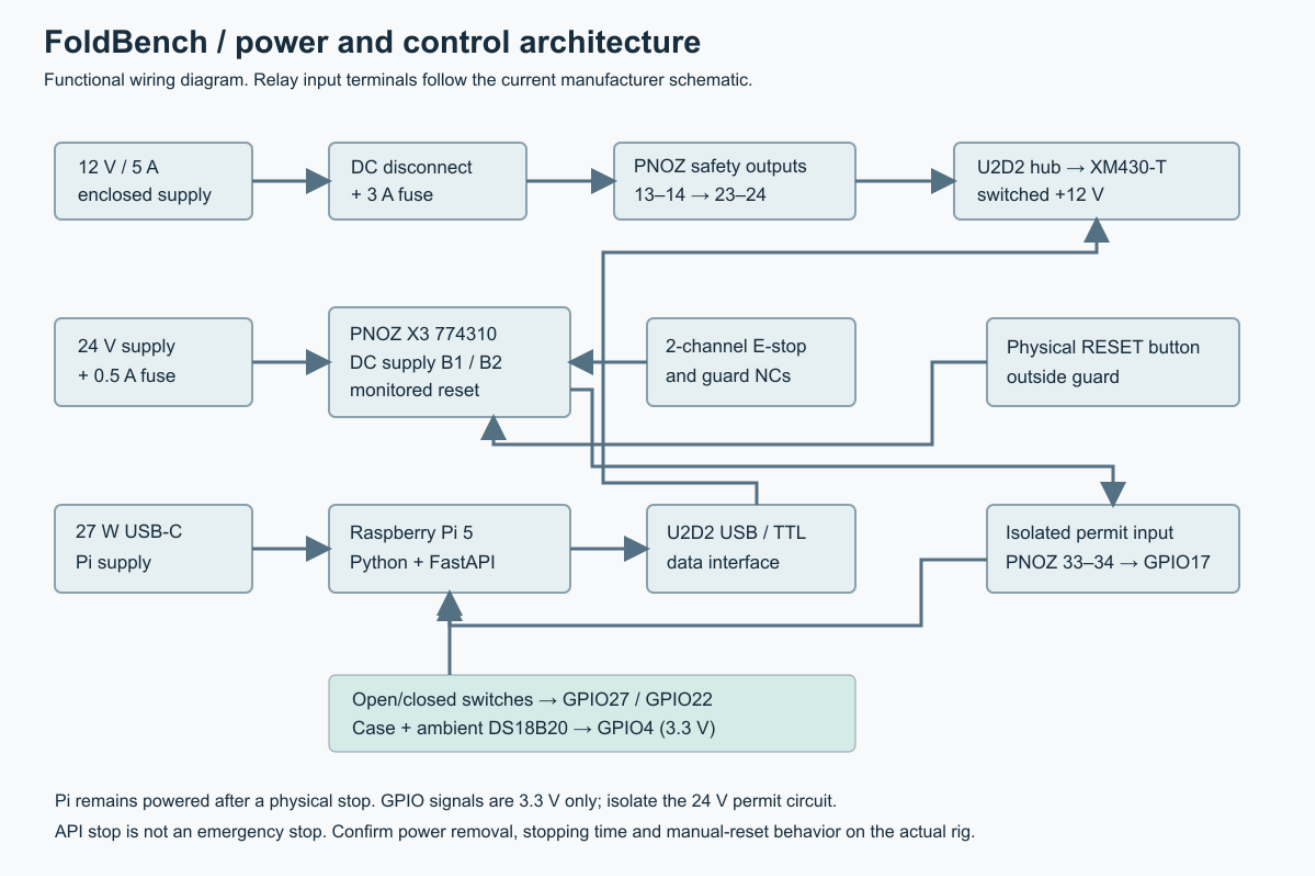
Software tests and generated schema were checked locally. No Pi, servo, phone or fabricated mechanism was connected during this work. Physical verification remains the commissioning work described above.

Assembly and wiring drawings



Editable vector originals and the OpenSCAD envelope model are in the hardware directory. Vendor mounting patterns and actual phone-contact profiles must be verified before fabrication.

Power and signal architecture



Use the detailed netlist and the manufacturer's current relay input schematic together. This drawing is a functional architecture, not a substitute for terminal-level safety-circuit verification.

Bill of materials - one FoldBench station

USD planning allowances, 2026-11-04. Verify quotes, interfaces and availability before ordering. Phone, taxes, shipping, machine-shop labor and workshop machinery are excluded. Fastener quantities include spares.

Station parts: $2,510.00. Tools if not owned: $415.00. Optional observation/charging: $115.00. Add a 25% station-parts contingency ($627.50) for a planning total of $3,137.50 before tools, optional items and labor.

Control - $143.00

ID	Qty	Item / specification	Unit USD	Ext. USD	Source
C01	1	Raspberry Pi 5: 4 GB; 64-bit Raspberry Pi OS	70.00	70.00	Vendor/reference
C02	1	Pi 27 W USB-C supply: Official supply; regional mains plug	15.00	15.00	Vendor/reference
C03	1	Active cooler and vented case: Pi 5 compatible; outside moving envelope	25.00	25.00	Vendor/reference
C04	1	64 GB high-endurance microSD: Genuine industrial/high-endurance card; retain backup image	25.00	25.00	Authorized memory distributor
C05	1	Ethernet cable: Cat6, 2 m	8.00	8.00	Electrical distributor

Drive - $904.00

ID	Qty	Item / specification	Unit USD	Ext. USD	Source
D01	1	XM430-W350-T actuator: ROBOTIS TTL variant; firmware 38+; ID 1; 57600 baud	330.00	330.00	Vendor/reference
D02	1	U2D2: ROBOTIS USB to TTL/RS-485 interface; use TTL	50.00	50.00	Vendor/reference
D03	1	U2D2 power hub board set: 12 V motor power distribution; keyed 3-pin TTL cables	25.00	25.00	Vendor/reference
D04	1	X430 horn/bracket set: Compatible metal horn, motor bracket and vendor screws; verify package contents	35.00	35.00	Vendor/reference
D05	1	USB cable: Pi USB-A to U2D2 connector, 0.5 m; match delivered interface	8.00	8.00	ROBOTIS or cable distributor
D06	2	DYNAMIXEL TTL cable: 3-pin keyed X-series, 200-300 mm, match delivered hub	8.00	16.00	Vendor/reference
D07	1	Slip-Ease EAS12 configured clutch: Request EAS12-6MM-6MM; horizontal; vendor-confirmed bidirectional 0.20-0.50 N m setting; quote required	220.00	220.00	Vendor/reference
D08	2	6 mm bearing pillow block: Ball bearing; split shaft support; rated >=50 N radial; vendor mounting drawing	20.00	40.00	Bearing supplier; select matched pair
D09	1	6 mm ground shaft stock: 300 mm stock cut into front/rear stubs and input link; no shaft through phone	20.00	20.00	Shaft supplier; lengths matched to clutch/support layout
D10	6	6 mm split shaft collar: Two per support plus input/assembly retention as needed	5.00	30.00	Shaft supplier
D11	2	6 mm split-clamp arm hub: Bolt-on flange, >=1 N m rating; match M03 arms	15.00	30.00	Mechanical distributor
D12	2	MGN7 linear guide with carriage: 100 mm rail; two matched rails; limited to +/-6 mm working float	30.00	60.00	Linear motion supplier; inspect free movement
D13	1	Adjustable friction damper: Rotary/hinge damper, bidirectional; qualify gravity settling; no claimed safety rating	40.00	40.00	Mechanical supplier; select from measured fixture inertia

Structure - $334.00

ID	Qty	Item / specification	Unit USD	Ext. USD	Source
M01	1	Base plate: 6061-T6 500 x 400 x 8 mm; drawing M01	100.00	100.00	Metal supplier / machine shop
M02	1	Fabricated plate stock set: All M02-M07 and M09-M13 parts in design table; machining labor excluded	130.00	130.00	Machine shop; dimension/slot schedule in guide
M03	8	Phone shoe blank: Acetal 20 x 15 x 8 mm; machine actual phone contours; M08	4.00	32.00	Plastics supplier / machine shop
M04	1	Silicone pad sheet: 200 x 200 x 1 mm, Shore A 30-40; removable adhesive if compatible	15.00	15.00	Industrial silicone supplier
M05	1	Compliant mount kit: 4 M4 shoulder screws, spacers, 8 silicone bushings, retaining washers; +/-2 mm travel	25.00	25.00	Mechanical distributor
M06	2	Adjustable mechanical stop: M6 screw, locknut, polyurethane bumper and M10 bracket	8.00	16.00	Mechanical distributor
M07	4	Rubber foot: M6 mounting, >=5 kg rating each	4.00	16.00	Mechanical distributor

Guard - $315.00

ID	Qty	Item / specification	Unit USD	Ext. USD	Source
G01	1	2020 aluminum extrusion frame: 4 x 560, 4 x 460, 4 x 360 mm plus base outriggers; 6 m stock allowance	90.00	90.00	Extrusion supplier; same slot series throughout
G02	24	2020 corner bracket: Matching M5 T-nuts and screws included	2.50	60.00	Extrusion supplier
G03	1	Polycarbonate panel set: Five 4 mm panels to M14 dimensions; allow cutting waste	120.00	120.00	Plastics supplier
G04	1	Door hardware set: 2 hinges, handle, latch, 24 panel clips and captive fasteners	45.00	45.00	Extrusion supplier

Electrical - $627.00

ID	Qty	Item / specification	Unit USD	Ext. USD	Source
E01	1	Enclosed 12 V DC supply: 5 A, SELV, listed brick, regional AC cord; matching locking DC inlet	35.00	35.00	Authorized power supply distributor
E02	1	Enclosed 24 V DC supply: 1 A or greater, SELV, listed brick and regional AC cord	30.00	30.00	Authorized power supply distributor
E03	1	PNOZ X3 774310: 24 V AC/DC safety relay; use B1/B2 for DC supply	300.00	300.00	Vendor/reference
E04	1	Emergency-stop assembly: 40 mm red mushroom/yellow background, twist release, 2 direct-opening NC contacts, 22 mm panel	45.00	45.00	Safety-component distributor; certified contact assembly
E05	1	OMRON D4NS-1BF guard switch: 2 NC direct-opening, tongue operated; matching PG13.5 gland	60.00	60.00	Vendor/reference
E06	1	OMRON D4DS-K2 actuator key: Match switch and chosen door entry direction; captive mounting	15.00	15.00	Vendor/reference
E07	1	Reset pushbutton: Momentary NO, 22 mm, blue label; physical monitored reset only	12.00	12.00	Industrial controls distributor
E08	1	DC disconnect: 2 pole, rated >=24 V DC 5 A; isolate low-voltage equipment for service	15.00	15.00	Electrical distributor
E09	1	Fuse/holder kit: 3 A fast DC F1, 0.5 A DC F2, covered holders plus 2 spares each; verify inrush and relay coordination	20.00	20.00	Electrical distributor
E10	1	Covered electrical enclosure: 300 x 250 x 150 mm nominal; mounting plate; outside guard sweep	60.00	60.00	Electrical enclosure supplier
E11	1	DIN/terminal kit: 300 mm DIN rail, end stops, 20 terminals, 10 jumpers, labels	35.00	35.00	Electrical distributor

Sensing - $52.00

ID	Qty	Item / specification	Unit USD	Ext. USD	Source
S01	2	OMRON D2F-01L endpoint switch: Gold-contact low-load SPDT; use COM/NO; adjustable output-side cams	5.00	10.00	Vendor/reference
S02	2	DS18B20 probe: Genuine 3-wire powered probe; 3.3 V; one case and one ambient	10.00	20.00	Authorized semiconductor/sensor distributor
S03	1	Isolated permit board parts: LTV-817C, 2 kohm 0.5 W, 1N4148, 3 x 10 kohm, 3 x 1 kohm, 4.7 kohm, covered perfboard and terminals	12.00	12.00	Electronics distributor; verify optocoupler pinout
S04	1	Probe attachment kit: Thin removable thermal tape, cable anchors, ambient standoff	10.00	10.00	Electronics distributor

Wiring - $75.00

ID	Qty	Item / specification	Unit USD	Ext. USD	Source
W01	1	18 AWG wire: Red/black, 3 m each, stranded, appropriately rated insulation	15.00	15.00	Electrical distributor
W02	1	Signal/safety wire set: 22 AWG 4-core 5 m; 24-26 AWG twisted pair 5 m	20.00	20.00	Electrical distributor
W03	1	Terminations and strain relief: 100 ferrules, 8 glands, 20 heat-shrink pieces, 20 tie bases, 50 ties	25.00	25.00	Electrical distributor
W04	1	Pi header breakout: 40-pin cable/breakout with screw terminals and standoffs	15.00	15.00	Electronics distributor

Fasteners - $60.00

ID	Qty	Item / specification	Unit USD	Ext. USD	Source
F01	1	M3 kit: 40 M3x8, 20 M3x12, 40 washers, 40 locknuts	10.00	10.00	Fastener distributor
F02	1	M4 kit: 40 M4x10, 20 M4x16, 80 washers, 40 locknuts	12.00	12.00	Fastener distributor
F03	1	M5/M6 kit: 40 M5x10/T-nuts, 20 M6x16, 40 M6 washers, 20 M6 locknuts	18.00	18.00	Fastener distributor
F04	1	Consumables: Low-strength threadlocker, witness paint, 0.1/0.2/0.5 mm shims	20.00	20.00	Industrial supplier

Tools - $415.00

ID	Qty	Item / specification	Unit USD	Ext. USD	Source
T01	1	Digital caliper and angle gauge: 0.1 mm fit measurement; angle resolution <=0.5 degrees	70.00	70.00	Metrology supplier
T02	1	Force gauge and 100 mm arm: 0-20 N, resolution <=0.1 N; documented calibration; custom arm	180.00	180.00	Metrology supplier; verify accuracy for acceptance
T03	1	Multimeter: DC voltage, continuity, adequate probe insulation	40.00	40.00	Authorized instrument distributor
T04	1	Assembly tools: Hex keys, low-range torque screwdriver, drill/taps/deburr tool access	100.00	100.00	Workshop allowance; machine tools excluded
T05	1	Two-leaf phone dummy: Match measured envelope and hinge path sufficiently for clearance trials	25.00	25.00	Fabricate; no claim to reproduce phone hinge torque

Optional - $115.00

ID	Qty	Item / specification	Unit USD	Ext. USD	Source
O01	1	Observation camera and mount: USB UVC camera for external recording; no camera control in delivered API	70.00	70.00	Camera supplier
O02	1	Phone charger/cable: Genuine compatible charger; strain relief; charging is separate qualified test condition	45.00	45.00	Apple / authorized distributor

Software - $0.00

ID	Qty	Item / specification	Unit USD	Ext. USD	Source
SW01	1	Python control stack: FastAPI, Uvicorn, dynamixel-sdk; Pipenv lock; OS and Python 3.12	0.00	0.00	Delivered source; no paid license allowance

foldatron / FoldBench - iPhone Duo endurance tester

An original prototype inspired by CNET's Fold-A-Phone video. Includes hardware design, a complete procurement/fabrication BOM, assembly and commissioning instructions, Python Raspberry Pi controller, FastAPI REST API, generated OpenAPI 3.1 specification and automated tests.

foldatron is the repository name; `foldbench` is the Python package and service name. The project is licensed under the existing [MIT license](LICENSE).

Default mode is simulation. Hardware has not been physically validated. The design is for supervised commissioning and testing; read the guide before fabrication or connecting an actuator. A mechanically completed run is not an automatic phone-function pass.

Files

• `docs/DESIGN_AND_ASSEMBLY.md`: design rationale, dimensions, wiring and assembly.

• `hardware/BOM.md` and `hardware/BOM.json`: quantities, specifications, allowances, procurement links.

• `hardware/layout.svg`: dimensioned assembly overview.

• `hardware/wiring.svg`: power and signal architecture.

• `hardware/foldbench.scad`: parametric assembly envelope and plate templates.

• `hardware/calibration.example.json`: uncommissioned installation limits.

• `docs/COMMISSIONING.md`: physical acceptance checklist.

• `docs/API.md`: endpoint behavior, error semantics and pagination.

• `openapi.json`: schema generated directly from the application.

• `src/foldbench/`: installable Python application, real servo/GPIO/probe driver and simulator.

• `deploy/`: service and environment examples for a Raspberry Pi.

• `tests/`: simulation/API and register-level driver tests.

• `scripts/`: BOM/manual generation, live HTTP check, wheel check and ZIP packaging.

• [Printable manual](docs/FoldBench-design-and-assembly.pdf): design, BOM, assembly and operation in one document.

Source layout

Application modules live under `src/foldbench/`, with package metadata in `pyproject.toml`. Pipenv installs this local package in editable mode. Tests exercise the installed package; they do not add the checkout or `src` to Python's import path. The same `python -m foldbench` command works outside the repository once installed. Tests, documentation, hardware files and deployment examples remain outside the Python import package.

Run the simulator

Use Python 3.12 and Pipenv. Runtime dependencies remain FastAPI, Uvicorn and the ROBOTIS SDK, with their transitive dependencies. Packaging, test and document-generation tools are development-only. Versions and hashes are in Pipfile.lock. No GPIO package is required: the real driver uses Linux's GPIO character-device ABI.

From this directory:

```
PIPENV_VENV_IN_PROJECT=1 pipenv sync --dev
export FOLDBENCH_API_KEY="$(pipenv run python -c 'import secrets; print(secrets.token_urlsafe(32))')"
export FOLDBENCH_MODE=simulation
pipenv run test
pipenv run serve
```

Open http://127.0.0.1:8000/docs and authorize with your generated `X-API-Key`. Use `GET /v1/status`, then `POST /v1/runs` with a unique `Idempotency-Key` and a body such as `{"specimen":"Duo-demo","cycles":3,"dwell_s":1}`. Default real-time simulation takes roughly 75 seconds for three cycles. Inspect the run, samples and CSV through the documentation UI. Stop the server with Ctrl-C.

`GET /health` is liveness only. Readiness and faults are in `/v1/status`. The API is intentionally bound to localhost by default. For remote access, use an SSH tunnel, or set a specific private LAN bind address with TLS configured. Keep the API off the public internet. There is no deployment to your Pi in this delivery: no Pi connection was supplied.

Raspberry Pi installation

Use Pi 5, 4 GB, active cooling, official 27 W supply, 64-bit Raspberry Pi OS, a high-endurance card or SSD, and Python 3.12. Install Pipenv with the OS package manager or your existing tool manager. If the OS supplies a different Python version, install 3.12 before syncing this locked environment, or deliberately re-lock and re-test for the chosen version.

Copy the application to `/opt/foldbench`, create the `foldbench` service account, create `/var/lib/foldbench` owned by it, and build the virtual environment there with Pipenv. Grant that account access to `dialout` and `gpio`, not unrestricted root access. Confirm the serial port's persistent `/dev/serial/by-id/` name and GPIO chip/header mapping. Enable 1-Wire on GPIO4 and record both probe serial numbers.

Copy the environment example to `/etc/foldbench.env`, set a unique key, actual device paths, and the absolute calibration path. Restrict the file to root and the service group. Use the supplied systemd unit after completing the physical checks. Run exactly one worker and no reload mode. A failed service restart never resumes an old test.

The hardware driver expects XM430-W350-T ID 1 at 57600 baud, protocol 2, firmware 38+. It requires independent endpoint switches, safety-permit input and both temperature probes. Missing hardware causes a fault; the program never silently substitutes simulation.

Validation

Run `pipenv run test`, `pipenv run lint`, and `pipenv run schema`. Compare a regenerated `openapi.json` with the distributed copy after API changes. Review `docs/VALIDATION.md` for the checks performed and the physical checks still outstanding.

`pipenv run package` builds a wheel and source archive. `pipenv run python scripts/check_wheel.py` verifies that the wheel imports and builds the API schema outside the source checkout. `pipenv run smoke` starts a temporary localhost server, completes a simulated cycle, checks the schema and shuts down. GitHub Actions runs these checks on pushes and pull requests.

Rebuild design deliverables

Edit `hardware/BOM.json`, then run `pipenv run bom` to update totals and the readable table. Run `pipenv run manual` to rebuild the PDF from the repository documentation and `docs/assets/` images. The generator uses fonts distributed with ReportLab and has no machine-specific font paths. If an SVG changes, regenerate its matching PNG in `docs/assets/` using `rsvg-convert` or another SVG renderer before rebuilding the manual, then visually review the PDF.

`pipenv run bundle` creates `artifacts/FoldBench-complete-package.zip` from Git-tracked files, including the PDF. Stage new files before bundling. GitHub's Code > Download ZIP also provides the complete repository. Virtual environments, credentials, test databases, caches and generated distribution archives are excluded from version control.

The code logs motor-derived angle and current as observations, not calibrated hinge torque or display-health measurements. Complete the commissioning checklist with an empty rig and dummy before testing the phone.

REST API contract

Base URL: `http://127.0.0.1:8000` by default. Use HTTPS or an SSH tunnel for remote clients. API key header: `X-API-Key`. Only health, interactive documentation and schema are public; they expose no run data or control actions.

Method	Path	Purpose
GET	/health	Process liveness, not equipment readiness
GET	/v1/status	Controller state, last/current run, last valid telemetry
GET	/v1/calibration	Read-only installation/commissioning limits
GET	/v1/telemetry	Latest observation; 503 when unavailable
POST	/v1/runs	Start a validated recipe; requires Idempotency-Key
GET	/v1/runs	Paginated history, limit and offset
GET	/v1/runs/{id}	Run state, recipe, calibration, confirmed count
POST	/v1/runs/{id}/pause	Request pause at the next completed cycle
POST	/v1/runs/{id}/resume	Resume a paused, inspected run
POST	/v1/runs/{id}/stop	End the active run and disable torque
POST	/v1/runs/{id}/inspections	Record pass/fail and notes while paused
POST	/v1/jogs	Move to a commissioned open/closed endpoint at 5 deg/s
POST	/v1/stop	Latched protective-stop request, including jogs
POST	/v1/reset	Acknowledge a resolved fault; no motion
GET	/v1/runs/{id}/samples	Sample page, after sequence and limit
GET	/v1/runs/{id}/samples.csv	CSV page, X-Next-After response cursor
GET	/v1/runs/{id}/events	Event/inspection page, after sequence and limit
GET	/openapi.json	Generated OpenAPI 3.1 specification
GET	/docs	Interactive API documentation

Start recipe

Required field: `specimen` (1-100 characters). Optional fields: `cycles` (integer 1-10,000,000; default 100), `speed_deg_s` (1-30; default 15), `dwell_s` (0.25-60; default 1), `current_limit_ma` (20-350; default 150), `inspection_every` (1-10,000; default 1,000). Local calibrated limits can be tighter and take precedence. Unknown fields and nonfinite numeric values are rejected.

The start request's `Idempotency-Key` must be 8-128 letters/digits/underscores/dots/hyphens. Use a new UUID for a deliberately new run. Repeating the same key and recipe returns the same run, including after restart; changing the recipe with that key returns 409. Never retry an uncertain start with a new key until reading status/history.

The run begins by confirming the open endpoint; that initial opening is not a cycle. A cycle is counted only after a subsequent confirmed closed endpoint and return to open. Each endpoint requires both a motor encoder tolerance and its independent switch, stable for 150 ms. Both switches high, or a switch active far from its calibrated encoder position, faults the run. Mode is always explicitly `simulation` or `hardware` in records.

State semantics

Controller states: idle, running, paused, jogging, faulted. Historical run states: running, paused, completed, stopped, faulted, interrupted. `active_run` in status contains the most recent run, including its terminal state, so it can be inspected after completion. Check controller state to determine ownership/readiness.

A pause request completes the current cycle and then removes torque. Automatic inspection pauses require a `pass` record before resume. A `fail` inspection faults the run. Resume reestablishes open position before beginning the next cycle; it does not increment a partial cycle. A terminal stopped/faulted/interrupted run cannot be resumed; start a new run after inspection and any required reset.

`POST /v1/stop` returns 202 immediately after requesting a latched stop. Poll status for the actual outcome. It is not an emergency-stop device. If a bus or controller call hangs, use the physical E-stop. A bus-watchdog fault requires inspecting the mechanism and cycling motor power before reset. Hardware reset does not bypass guard, temperature, position or other readiness checks.

Records and export

Samples are normally stored once per second plus endpoint observations; timestamps are UTC. Motor current is signed mA, angle is estimated from the motor encoder and local calibration, and temperatures are Celsius. Endpoint sensors verify position at endpoints only. Motor angle can diverge from phone angle between endpoints if the clutch slips or a clamp moves.

For complete JSON/CSV export, request pages until an empty page: `limit` defaults to 1,000, maximum 10,000; the next `after` is the last returned `seq`, or CSV's `X-Next-After`. CSV repeats its header per page. Sequence numbers are database-global, so gaps within a run are normal. Downloading during a run produces an evolving dataset; finish/stop first for a stable export. Results remain local in SQLite; no cloud upload is performed.

A crash after physical motion but before a confirmed-cycle commit may undercount the last cycle. On startup, all previously running/paused records are marked interrupted and retain their committed count. Backup the database with SQLite's backup facility, not a bare copy of a live database that omits its WAL.

HTTP responses

200: read/control result. 201: created or idempotently replayed start. 202: asynchronous stop/jog request accepted. 401: missing/incorrect key. 404: unknown run. 409: ownership, calibration, safety state or controller conflict. 422: invalid input. 503: no valid telemetry. Detailed schemas and response models are in openapi.json. Do not interpret HTTP 200/201/202 alone as a mechanical pass or proof of zero motion.

Commissioning record

Record ID: _____ Date: _____ Operator/reviewer: _____

Station serial/revision: _____ Servo serial/firmware: _____

Phone serial/iOS: _____ Fixture shoe revision: _____

This checklist is a physical test record. Software simulation results do not fill these boxes.

Check	Required evidence	Result / initials
Actual phone envelope	Half widths, height, thickness, camera/button/port map	
Hinge migration	Radial/normal position at five angles; >=2 mm spare travel	
Cradle retention	10 N tangential dummy test, no phone/screen damage	
Axis alignment	Bearing and clutch alignment measured	
Unpowered clearance	Entire swept volume clear; no hinge shaft intrusion	
Clutch	Bidirectional breakaway torque, repeatability, lock mark	
Baseline hinge load	Force/radius/direction/angle and empty-fixture load	
Hard stops	Measured endpoints within specimen free range	
Endpoint switches	Independent output-side operation and disagreement fault	
12 V polarity/fuse	Meter reading and fuse identification	
24 V relay wiring	Vendor circuit revision and terminal inspection	
GPIO isolation	Permit high/low voltage; no 24 V on Pi side	
E-stop	Motor power removed independently of Pi; manual reset required	
Guard	Each contact/channel tested; no restart on closing alone	
Stop distance/time	Worst motion and gravity settling; reach-in prevented	
USB disconnected	Watchdog stops; no automatic resumption	
Pi process/power loss	Motion response and restart behavior	
Motor power cycle	Torque stays off until explicit valid command	
Probe mapping	Case/ambient serials, CRC failure, disconnected-probe fault	
Probe thermal contact	Case attachment retained through complete arc	
Dummy jam/slip	Fault, no false increment, clutch setting retained	
100 dummy cycles	Independent count agreement and no wear/movement	
1,000 dummy cycles	Temperatures and drive/fixture duty validated	
Phone single/10/100 cycles	Functional inspection results at each stage	

Check	Required evidence	Result / initials
Network access	Unique key, private/TLS or tunnel, one worker only	
Disk/restart	Records durable, interrupted runs never resume	

Calibrated closed tick/angle: _____ Open tick/angle: _____

Maximum speed/current: _____ Clutch torque: _____

Case/ambient/motor temperature thresholds: _____

Acceptance disposition and remaining restrictions: _____

Release for supervised phone trials: _____

Separate release for any unattended testing: _____

Validation evidence and limits

Date: 4 November 2026. Execution host: local macOS, Python 3.12.14, Pipenv-managed environment. No physical Raspberry Pi, servo or phone was connected.

Passed locally

• 25 automated tests: authenticated API access; request bounds; generated security schema; durable complete-cycle count; CSV/JSON pagination; idempotent retries; exclusive run ownership; concurrent starts; latched stop; no partial-cycle increment; permit/temperature/current faults; inspection pass/resume and fail termination; manual pause/stop; endpoint jog; jam timeout; stuck sensor; crash recovery; process lock; missing-key rejection; probe CRC/range; database failure; servo register write ordering; commissioned-limit checks; communication errors; rejected illegal goals; uncoupled index current/speed limits, permit gating and final torque disable.

• Ruff style/import/error checks, configured for PEP 8-compatible 79-character lines.

• A real local Uvicorn HTTP run: authenticated POST, status polling, completed simulated cycle, saved run result and exact equality between live and exported OpenAPI schemas.

• Dependency resolution and installation through Pipenv; committed-to-deliverable lockfile records package versions and hashes.

• Source-layout migration: `src/foldbench` installed through Pipenv in editable mode, with no pytest import-path override; wheel and source distributions built, and wheel import/API schema checked from outside the checkout.

• Assembly and wiring SVGs rendered for visual inspection. OpenSCAD source supplied as an envelope/plate-template model; no OpenSCAD executable was available for mesh rendering or collision verification.

The test runner reports one upstream Starlette warning about future httpx test-client deprecation. It does not prevent the current tests from passing. The delivered dependencies remain locked; review this warning when upgrading them.

Not established by these checks

GPIO operation on the selected Pi/kernel, 1-Wire timing with actual probes, DYNAMIXEL serial behavior under real electrical faults, actual clutch torque, alignment, phone clearance, stopping distance, fixture wear, cycle-counter accuracy under physical slip, hardware temperature thresholds, continuous-duty endurance, safety-circuit performance and handset functional health remain unverified. Driver register tests use a fake transport and cannot validate actuator behavior.

The setup indexing utility must be exercised only with the motor mechanically uncoupled and no phone. Its physical effect is unverified. The software's local commissioning record is a lockout mechanism, not proof that an engineer performed the checks.

See COMMISSIONING.md for the physical evidence required before phone trials. No deployment was made to a Raspberry Pi, and the temporary HTTP smoke-test process was stopped after verification.